DATA MANIPULATION LANGUAGE (DML) :

IT IS USED TO MANIPULATE THE OBJECT / TABLE BY PERFORMING OPERATIONS LIKE INSERTION, UPDATION AND DELETION.

TYPES OF DML STATEMENTS

1. INSERT
2. UPDATE
3. DELETE

NOT AUTO COMMIT
NOT AUTO SAVED

› INSERT

IT IS USED TO CREATE / INSERT THE RECORDS IN THE TABLE

SYNTAX:

```
INSERT INTO TABLE_NAME VALUES (V1, V2, V3..... Vn);
```

INSERT MULTIPLE RECORDS

```
INSERT INTO TABLE_NAME VALUES (&COL1, &COL2..... &COLn);
```

› UPDATE

IT IS USED TO MODIFY / UPDATE THE EXISTING VALUE

SYNTAX:

```
UPDATE TABLE_NAME
SET COL_NAME = VALUE , COL2_NAME = VALUE...
 [WHERE < FILTER CONDITION >]  ;
```

› DELETE

IT IS USED TO REMOVE / DELETE A PARTICULAR RECORD FROM THE TABLE

SYNTAX:

```
DELETE  FROM TABLE_NAME
[WHERE < FILTER CONDITION >] ;
```

TRANSACTION CONTROL LANGUAGE (TCL) :

IT IS USED TO CONTROL THE TRANSACTION ON THE DATABASE.

THE OPERATIONS PERFORMED ON DML

TYPES OF TCL STATEMENTS

1. COMMIT
2. ROLLBACK
3. SAVEPOINT

› COMMIT

IT IS USED TO SAVE THE RECORDS / TRANSACTIONS INTO DATABASE

SYNTAX:

```
COMMIT;
```

› ROLLBACK

IT IS USED TO GET BACK TO THE LATEST SAVED LOCATION

SYNTAX:

```
ROLLBACK;
```

EX:

		INSERT	DISPLAY	ROLLBACK;
	1	1	1	1
	2	2	2	2
	3	COMMIT;	3	3
DATA -->	4	3	4	4
	5	4	6	INSERT 5
	6	COMMIT;	7	6
	7	6	FORGOT TO INSERT 5 REC	7 DELETED
		7		

› SAVEPOINT

IT IS USED TO GIVE RESTORATION POINTS / MARK POSITION

SYNTAX:

```
SAVEPOINT SAVEPOINT_NAME;
```

EX:

		INSERT	DISPLAY	ROLLBACK;
	1	1	1	1
	2	2	2	2
	3	COMMIT;	3	3
DATA -->	4	3	4	4
	5	4	6	6
	6	COMMIT;	7	7
	7	6	9	9
	8	7	FORGOT TO INSERT 5,8 REC	SAVED TILL 9TH
	9	COMMIT;		
		9		
	ROLLBACK;	COMMIT;		

EX:

		INSERT	DISPLAY
	1	1	1
	2	2	2
	3	SAVEPOINT S1;	3
DATA -->	4	3	4
	5	4	6
	6	SAVEPOINT S2;	7
	7	6	9
	8	7	FORGOT TO INSERT 5,8 REC
	9	SAVEPOINT S3;	
		9	
		SAVEPOINT S4;	

--->

INSERT 8	DISPLAY	INSERT 5
ROLLBACK TO S3;	1	ROLLBACK TO S2;
1	2	1
2	3	2
SAVEPOINT S1;	4	SAVEPOINT S1;
3	6	3
4	7	4
SAVEPOINT S2;	8	SAVEPOINT S2;
6	FORGOT TO INSERT 5 REC	INSERT 5
7		6
SAVEPOINT S3;		7 DELETED
INSERT 8		8
9 DELETED		

> DATA CONTROL LANGUAGE (DCL) :

IT IS USED TO CONTROL THE FLOW BETWEEN THE USERS.

TYPES OF DCL STATEMENTS

1. GRANT
2. REVOKE

GRANT :

IT IS USED TO GIVE PERMISSION TO USER

SYNTAX:

```
GRANT SQL_STATEMENTS
ON TABLE_NAME
TO USER_NAME;
```

CONNECT

USER_NAME :
PASSWORD : tiger

SHOW USER;

SELECT *
FROM SCOTT.EMP;

EX:

```
GRANT SELECT
ON EMP
TO HR;
```

REVOKE :

IT IS USED TAKE BACK THE GIVE PERMISSION TO USER

SYNTAX:

```
REVOKE SQL_STATEMENTS
ON TABLE_NAME
FROM USER_NAME;
```

EX:

```
REVOKE SELECT
ON EMP
FROM HR;
```

DIFFERENCES BETWEEN DROP ,TRUNCATE, DELETE